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Use of decision tree models based on evolutionary algorithms for the morphological classification of reinforcing nano-particle aggregates

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ABSTRACT

An original method is presented to classify nano-particle aggregates into one of the morphological classes that were previously proposed in the literature. Carbon black was selected as a study case in this work, since it is a nanoreinforcement used massively in many industries, but the proposed method can be applied to other particulate aggregates as well. This methodology consists of three steps: (i) transmission electron microscopy image processing to compute a set of twenty-one morphological characteristics for each aggregate including the new attributes proposed herein, (ii) a multivariate analysis of the dataset to reduce the problem dimensionality and (iii) creation and assessment of decision trees based on evolutionary algorithms to classify the aggregates. The effectiveness of the method was proven for a balanced-per-class sample of forty-eight selected aggregates. The original dimension of the problem is reduced to three principal components, which explain 90% of the total variance. The best model classifies an aggregate into one of the four morphological classes in a simple comparison process. The accuracy of the applied model to classify new aggregates was 75%.

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1. Introduction

Many properties of composite materials depend on the individual properties of each constituent, the interaction among the constituents and the morphology of the composite. Hence, to understand and eventually predict the final properties of a material, a good characterization of the morphology is necessary.

Rubber blends are usually reinforced with nano-particles (mainly carbon black or silica). These fillers consist of spherical particles that are partially fused together to form aggregates, and those aggregates agglomerate because of Van der Waals forces [1]. The fillers can have different shapes, which are usually classified into four categories [2]: spheroidal, ellipsoidal, linear and branched, as shown in Fig. 1. The size and shape of these filler particles significantly affect the final properties of the rubber blend [3].

For the carbon black filler, the size and shape information of the particles and the occurrence of each shape are contained in a filler characteristic, which is ambiguously named "structure" and includes the information of the particle size and the specific surface area that is accessible to the rubber. The structure can be measured using both indirect and direct methods. Indirect methods are bulk colloidal methods that provide averaged results and are mostly performed in industrial laboratories following standard procedures [4,5]. These techniques are easy to apply and report quick and useful results that allow one to compare different grades of carbon black. Direct methods are based on transmission electron microscopy (TEM), which is used to visualize and characterize individual carbon black aggregates. The corresponding ASTM standard defines the procedure for sample preparation, image processing and aggregate data extraction [6]. The calculations that are specified in the standard must be applied to thousands of aggregates, and an averaged morphological characterization is finally obtained. Among the geometrical characteristics, when the microscope is set to the proper magnification, the electron microscope surface area (EMSA) is calculated, and its value is similar to the values obtained using colloidal methods (STSA [5]). The direct methodology improves the gain in the filler structure information because it provides distributional information for the data. However, the efficient quantification of the occurrence of different morphological classes and the ratios among them remains unsolved.

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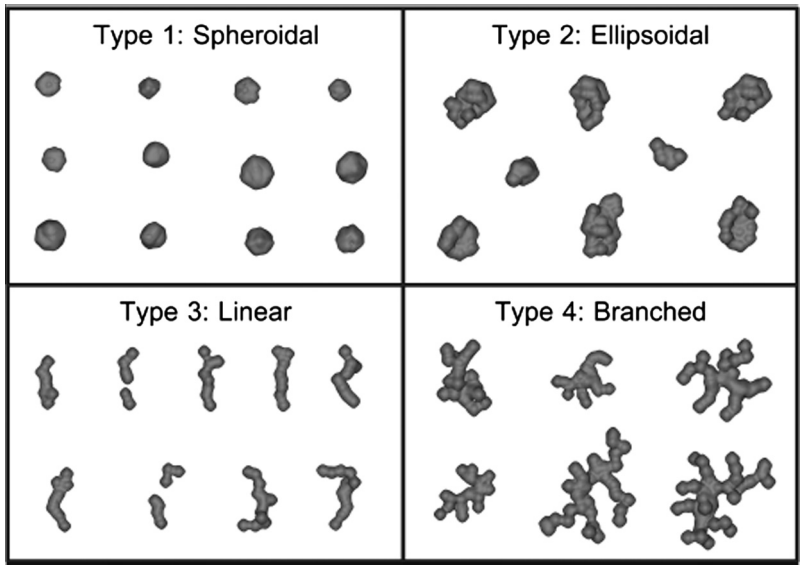

Fig. 1. Morphological categories of carbon black aggregates (from Ref. [13]).

To define the appropriate aggregate categories and the methodology to successfully and efficiently classify an object has been an issue of interest in the last decades. Image analysis was first used by Medalia to analyze the shape and size of carbon black aggregates by manually digitalizing them from TEM micrographs [7]. The shape classification proposed by Medalia was based on only a few geometrical parameters. The efficiency of this technique was significantly improved by using digital cameras to combine TEM and Automated Image Analysis (AIA) [8]. The first computerized AIA analyses of carbon black TEM images were performed by Hess and co-workers in the late sixties and early seventies [9,10]. They presented a more advanced shape-classification method based on additional non-dimensional form factors, which were calculated from the morphological data from the 2D projection of carbon black particles.

These classification efforts were later improved by Bourrat et al. by adding some new geometrical characteristics [11], but they were based purely on two-dimensional images (binarized images). Subsequently, Maas and Gronski used microdensitometry (an analysis of the grey level of the TEM images) to include three-dimensional structure information in the shape analysis [12].

In the methods proposed by Herd et al. [3] and Maas and Gronski [12], the final classification is performed using a number of discrimination/descriptor equations. Those equations are produced by statistically analyzing a number of chosen discrimination parameters in relation to the shape class to which the corresponding aggregate belongs. The selected parameters are the traditional size/shape characteristics, and besides several others that are derived from the skeletonization process and the fractal approach. These works do not report the accuracy of the classification methods. Recently, a new methodology of automatic classification was proposed by López-de-Uralde et al. [13]. They used machine learning algorithms with supervised training to classify the aggregates into Herd's four classes. The algorithms used more than forty characteristics, which were calculated from the projected area and the perimeter of the aggregates. The reported accuracy varied in the range of 49–84% depending on the applied algorithm and whether the sample was artificially balanced per class (by creating artificial spheroidal aggregates, which occur less frequently in carbon black samples).

Despite these efforts, the classification of an aggregate into a proposed category remains controversial, even if it is consistent with the categories. Man-made classification can result in different assignments for the same object, and the inconsistency is dramatically increased if the process is automatically implemented, i.e., if the TEM images are processed without careful human post-supervising. Details such as the illumination deficiencies or dirt in the images can lead to errors in the definition of the aggregate limits. Thus, although the aim is to measure the structure of carbon black grades using direct methods, i.e., by reporting the distribution of the morphological characteristics and classifying the aggregates according to their shapes, the solution remains a demanding problem.

Currently, with the great improvements in computing capacity and by applying pattern recognition and machine learning algorithms to this problem, one can exhaustively and properly analyze large amounts of information. Thus, the accuracy models can be implemented to automatically classify carbon black aggregates and consequently obtain complementary information about the filler structure. This work proposes a three-step method to provide enhanced and deeper knowledge of the filler characteristics for future modeling of filled-rubber properties.

In the first step, a set of twenty-one attributes is defined, measured and calculated using the TEM images of forty-eight selected aggregates – twelve per class – to obtain a balanced sample. In addition to the most common descriptors that are used to describe the size and shape of an object [14], the set includes for the first time Fourier descriptors and six new parameters, which are defined by the authors.

In the second step, multivariate analysis techniques are applied to analyze the data. This step combines three techniques: analysis of the variance homogeneity, correlation analysis and principal component analysis (PCA). The aim is to obtain a better understanding of the descriptor characteristics and their effects on the shape categories. With this analysis, because the number of initial attributes is dramatically reduced, it is easier to interpret and use the final classification model, which is proposed in the last step.

In the last step, a model is implemented based on decision trees to classify the shape of the carbon black aggregates into one of the four categories that were proposed by Herd. The trees that we used in this work are based on evtree algorithms, which search over the parameter space of trees using global optimization methods. This type of tree uses evolutionary algorithms that implement four types of mutation operators and one crossover operator; thus, in

many cases, it improves the results of heuristic trees such as CART or conditional inference trees. Despite the disadvantage of a higher computational cost, the final results of the evtree algorithms are improved [15–17]. The inputs for the models were the principal components obtained in the previous step. The outputs were assessed according to different criteria.

The method shows promising results. Because the original number of variables is reduced to only three principal components and the generated model can successfully classify the morphology, direct information of the carbon black structure can be obtained. The results also enhance the implementation of the carbon black structure information in modeling the filled-rubber properties [18].

2. Experimental

2.1. Materials, image acquisition and pre-treatment

To perform this work, carbon black of grade Vulcan XC605, which was supplied by Cabot Corporation, was used. The samples were prepared according to the instructions for dry powder in Ref. [6]. Images of the aggregates were captured on a Philips CM120 Biofilter transmission electron microscope, which operated at 120 keV with a pixel size of 2.34 nm. A set of forty-eight aggregates was selected as a representative sample of four morphological categories and includes a wide range of sizes for each class. This set of aggregates contains twelve aggregates per category to provide a balanced sample (Fig. 2). It is necessary to note that some of the branched aggregates can be agglomerates of few aggregates, but they were specifically selected because this type of object occurs with notable frequency when carbon black samples are prepared for the TEM analysis.

The images were individually processed to obtain the binarized silhouettes using the methodology that was proposed in the ASTM standard [6]. Noise was removed by applying an anisotropic diffusion filter; then, a threshold value was calculated using Otsu's method [19] and applied to obtain the binarized silhouette of the aggregates. Finally, the edges were smoothed. When the forty-eight images were successfully processed, each silhouette was isolated from the original image (Fig. 3), and the aggregates were characterized by measuring twenty-one parameters. Both the image processing and the silhouette characterization were implemented in Matlab using the Image Processing Toolbox 5.0.

2.2. Carbon black structure data

2.2.1. Geometrical and morphological characterizations

The classification of two-dimensional shapes regardless of their size, position and orientation, i.e., invariant attributes, is a problem in pattern recognition. In this work, both invariant and non-invariant morphometrical characteristics were studied. The silhouette of each aggregate was rotated to set its maximum Feret diameter [20] to the horizontal position. After the rotation, twenty-one attributes, which were proposed by the standard [6], related to the aggregate perimeter (i.e., the boundary) and the projected area (i.e., the region), were calculated for each aggregate.

2.2.1.1. Aggregate characteristics defined in ASTM Standard D3849-07. There are eight attributes in this group. The aggregate area (A) and the aggregate perimeter (P) are directly measured from the image. The diameter of a circle with the same area (D), the aggregation factor (α), the average particle size for an aggregate (d_p), the aggregate volume (V_A), the particle volume (V_p) and the number of particles in the aggregate (n) are derived from A and P.

2.2.1.2. Aggregate characteristics based on the boundary. In this subset, five attributes were calculated; the first three are extensively used in pattern recognition, and the other two are originally proposed in this work:

- Roughness (Ro): The ratio between the aggregate perimeter and the convex-hull perimeter, which wraps the aggregate.
- Signature (SSD): The signature of the boundary is a one-dimensional representation of the distance from the centroid to the boundary [21,22]. It is calculated using the standard deviation of 360 distances from the centroid to the boundary, one per degree where the distance is measured. For spheroidal aggregates, where the boundary is notably uniform, this value should be lower, whereas it should be higher for branched aggregates.
- Fourier descriptors (FD): Fourier transformation of the boundary signature has been popular for classification purposes. The method generates a complete set of complex numbers, the Fourier descriptors, which represent the shape of the object in a frequency domain [23–25]. The lower frequency descriptors contain information of the general shape, and the higher frequency descriptors contain information about the smaller

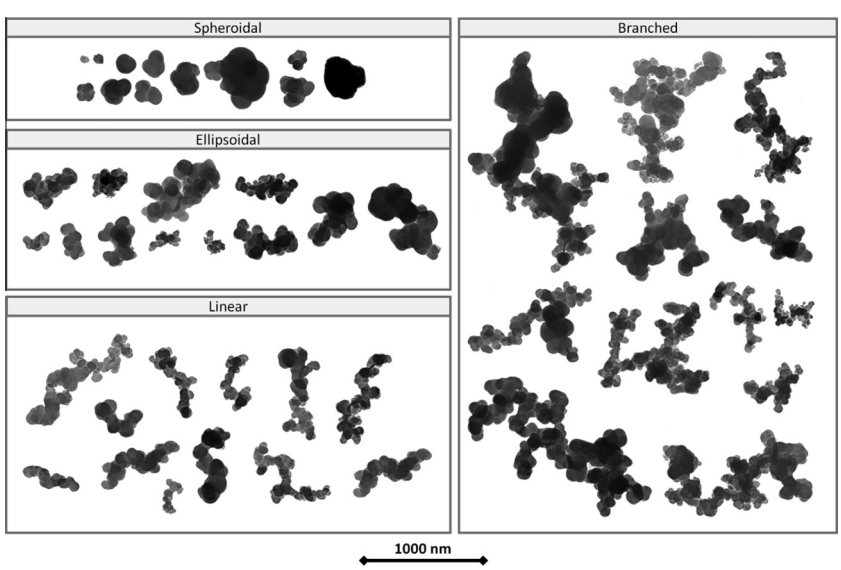

Fig. 2. TEM images of the 48 aggregates selected to perform this work. Aggregates are grouped into the four categories: spheroidal, ellipsoidal, linear and branched.

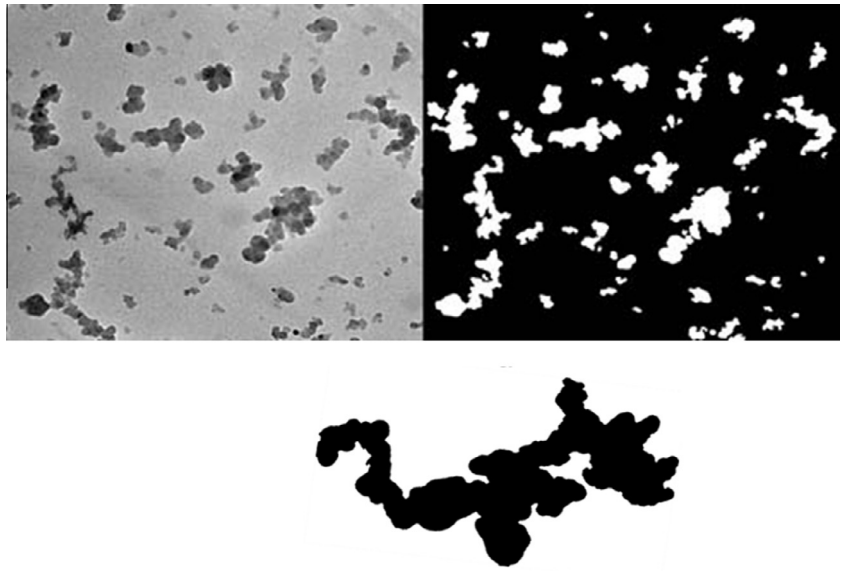

Fig. 3. Semi-automated TEM-image processing and aggregates isolation.

details. A subset of the Fourier descriptors is often sufficient to discriminate different shapes: the subset size depends on the irregularity of the shape and the desired details of the boundary. Thus, in this work, it was necessary to define the necessary number of Fourier descriptors to obtain a sufficiently good description of the aggregate shape that allows classifying. Hence, an error function was defined by the ratio of the summation of the distances between any original point on the boundary and its computed position, which is calculated from the inverse Fourier transform, to the total length of the original perimeter. A basic analysis of the preliminary tests determined that the number of Fourier descriptors that compute a value of the error function below 5% was a proper criterion (Fig. 4) for this classification.

To define two new attributes, one hundred segments were generated in the region of the aggregate as shown in Fig. 5a, and the number of intersections with the boundary was computed. The segment length and the starting and ending points are randomly defined with the unique restriction that both ends must fall inside the region covered by the aggregate. It is assumed that for regular shapes (spheroidal aggregates), only a reduced number of

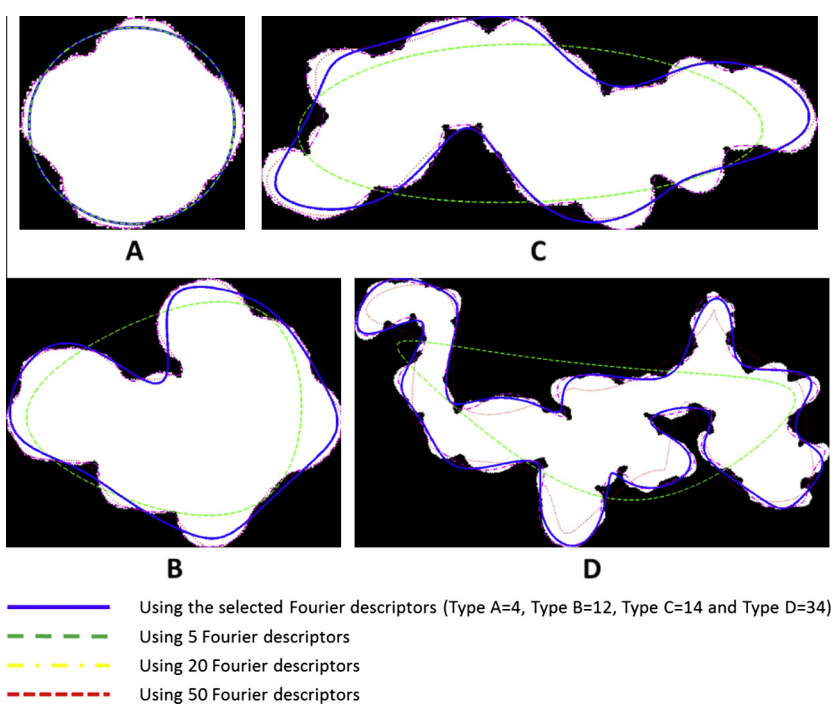

Fig. 4. Effect of the number of Fourier descriptors used to reconstruct the boundary of the aggregates. Blue lines draw the reconstructed boundary of each aggregate using the number of descriptors that satisfy the error criteria proposed in this work. This is number that defines the value of the attribute *FD* for each aggregate. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

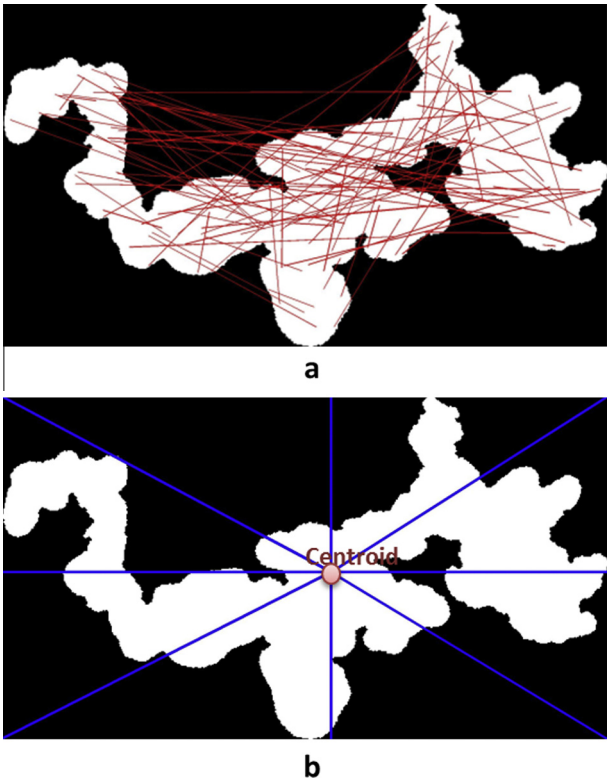

Fig. 5. Illustration of the attributes proposed to improve the classification of carbon black aggregates. (a) Measure of changes in concavity and convexity: red lines are random segments used to measure *CCseg* and *CCcut* attributes. (b) Division of the bounding box into the eight sectors for the discrete approach. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

segments should intercept the perimeter, and the total number of interceptions is low. For irregular shapes (branched aggregates), the number of segments that intercept the boundary and the total number of interceptions are high because a segment can cut the aggregate boundary more than once. Thus, the newly defined parameters are:

- The ratio between the number of segments that intercept the boundary and the total number of generated segments (*CCseg*).
- The total number of interceptions (*CCcu*).

2.2.1.3. Aggregate characteristics based on the region. In this subset, eight attributes were calculated. The first four are extensively used in pattern recognition, and the other four are derived from a discrete approach of two of them, which were computed to obtain more information about the distribution of the aggregate region inside its enclosing bounding box.

- Circularity (C): The roundness of the projected image of the aggregate is defined in terms of the circularity according to Ref. [26]. $C = 1$ corresponds to a perfect circle, C approaches 0 as the polygon elongates, and a line segment is the limit.

$$C = \frac{4\pi A}{P} \tag{1}$$

- Solidity (So): The ratio between the aggregate area and the area of the convex polygon that wraps the aggregate.
- Extent (Ex): The ratio between the aggregate area and the area of the bounding box of the aggregate.
- Eccentricity (Ec): The eccentricity of the ellipse that has the identical second moments as the region, which is the ratio of the distance between the foci of the ellipse to its major axis length. $Ec = 0$ corresponds to a perfect circle, and $Ec = 1$ corresponds to a line segment.

To discreetly compute the Extent and the Solidity, the bounding box rectangle was divided into eight triangles (Fig. 5b). The three vertices of each triangle were the centroid of the region, the bounding box vertices and the middle points of each bounding box side. After creating the eight subparts, the Solidity and the Extent of the new subregions were measured and evaluated using the following attributes.

- The mean and the standard deviation of the Extent that was applied to the eight subparts (*ExMean*, *ExSD*).
- The mean and the standard deviation of the Solidity that was applied to the eight subparts (*SoMean*, *SoSD*).

After the twenty-one attributes were measured and calculated, the dataset was normalized between 0 and 1 to balance the weight of all variables and improve the statistical-analysis quality.

2.2.2. Multivariate analysis of the data

The multivariate analysis of the gathered information was performed in three steps using the R suite [27] for the statistical analysis and subsequent modeling.

In the first step, a variance analysis was performed. For each variable, the technique computes evidence of significant differences among the classes. The aim is to detect the most useful variables that make the clearest differences and the non-useful variables that could increase the noise in the data, which reduces the accuracy of subsequent classifications models. Many statistics tests can be performed depending on whether the variable distribution is parametric or non-parametric, the shape of the non-Gaussian distributions (skewed and extremely heavy-tailed distributions), and the sensitivity to outliers or others. In this work, ANOVA [28,29], Bartlett [30], Brown–Forsythe [31,32] and Fligner–Killeen [33,34] tests were performed for twenty-one variables in the dataset. All of them check the null hypothesis, i.e., whether the homogeneity of variance occurs. When the null hypothesis is rejected, the variable distinguishes the classes; if the null hypothesis is satisfied, the variable is discarded from the dataset. The test results can be analyzed using two approaches: the classical method compares the statistics of the variable with the critical value, which is calculated for a given significance level (usually 95%) and the degree of freedom of the sample; the newer approach computes the p-value that corresponds to the statistics of the test and checks it against a given significance level (95% if the desired restrictive criterion is identical to that in the former approach).

In the second step, a correlation analysis was performed to identify the redundant variables that might not contribute to improve the model accuracy and could be removed to decrease the noise and reduce the problem dimension. The analyst must define the critical value of the correlation factor, which makes a variable redundant in the problem.

In the last step, PCA was applied to further reduce the dimension of the problem. The PCA is one of the most popular and successful methods for this propose [35,36]. The technique applies an orthogonal transformation of the initial space of variables, which creates a new set of variables that are named the principal components. The computed orthogonal projection captures the majority of the variance in the data and converts a set of possibly correlated variables into a new set of linearly uncorrelated

components [37–39]. This method, which is based on a linear transformation, provides the weight of the original variables on the new principal components.

2.2.3. Classification model: evolutionary learning of globally optimal classification trees

A classification tree is a tool to explore the relationship between a single nominal response variable and multiple explanatory variables [40]. The classification tree model is fitted using binary recursive partitioning, where the data are successively split along the coordinate axes of the explanatory variables.

Classical classification tree methods build the model in a forward stepwise search where the splits are chosen to maximize the homogeneity only in the next step. In this paper, global optimization methods based on evolutionary algorithms are used to search over the variable space; more specifically, an evtree algorithm (Evolutionary Learning of Globally Optimal Classification Trees in R) is used [15].

During the training process, a collection of trees are iteratively modified using the variation operators. Four types of mutation operators (split, prune, major split rule mutation and minor split rule mutation) and one crossover operator (which exchange the subtrees between two trees) are used in the evtree algorithm. The solution with the best performance is selected by taking the fitness function as the quality criterion. This function maximizes the model accuracy on the training data and minimizes the model complexity. In the model, a parameter α measures the complexity

Table 1
Metrics used to optimize the evtree algorithm based model.

$TP\ Rate = \frac{TP}{TP+FN}$	$FP\ Rate = \frac{FP}{FP+TN}$
$Precision = \frac{TP}{TP+FN}$	$Recall = \frac{TP}{TP+FN}$
$Fmeasure = 2 \cdot \frac{Precision \cdot Recall}{Precision+Recall}$	$Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)}$

TP: aggregate has the condition and tests positive for the condition.
TN: aggregate does not have the condition and tests negative for the condition.
FP: aggregate does not have the condition but tests positive for the condition.
FN: aggregate has the condition but tests negative for the condition.

Table 2
Descriptive statistics for the 21 original variables. Classes: A = spherical, B = ellipsoidal, C = linear and D = branched. IQR is the interquartile range.

Class	Attribute	Mean	sd	IQR	Min	25%	50%	75%	Max	Attribute	Mean	sd	IQR	Min	25%	50%	75%	Max
A	*A*	0.02	0.02	0.01	0.00	0.00	0.01	0.02	0.08	*P*	0.12	0.09	0.09	0.00	0.06	0.10	0.16	0.32
B		0.04	0.04	0.04	0.01	0.01	0.04	0.05	0.12		0.20	0.10	0.14	0.08	0.12	0.21	0.26	0.38
C		0.10	0.08	0.08	0.01	0.04	0.10	0.12	0.33		0.29	0.11	0.14	0.11	0.21	0.31	0.35	0.53
D		0.34	0.29	0.20	0.04	0.16	0.28	0.36	1.00		0.54	0.24	0.25	0.18	0.40	0.54	0.64	1.00
A	*D*	0.14	0.13	0.12	0.00	0.06	0.13	0.18	0.38	α	0.05	0.15	0.00	0.00	0.00	0.00	0.00	0.51
B		0.27	0.10	0.15	0.14	0.19	0.28	0.34	0.47		0.14	0.14	0.26	0.00	0.00	0.09	0.26	0.37
C		0.43	0.15	0.19	0.21	0.32	0.45	0.52	0.74		0.50	0.31	0.46	0.02	0.28	0.44	0.75	1.00
D		0.65	0.19	0.17	0.34	0.55	0.64	0.72	1.00		0.66	0.14	0.21	0.41	0.57	0.68	0.78	0.83
A	d_p	0.04	0.05	0.03	0.00	0.01	0.03	0.04	0.18	*Va*	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.02
B		0.09	0.05	0.07	0.03	0.05	0.08	0.12	0.22		0.01	0.01	0.01	0.00	0.00	0.01	0.01	0.04
C		0.24	0.17	0.22	0.06	0.12	0.21	0.34	0.65		0.04	0.05	0.03	0.00	0.01	0.03	0.04	0.20
D		0.48	0.25	0.21	0.16	0.35	0.38	0.56	1.00		0.24	0.30	0.15	0.01	0.06	0.14	0.21	1.00
A	V_p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	*n*	0.60	0.22	0.12	0.08	0.57	0.62	0.69	1.00
B		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01		0.35	0.16	0.29	0.13	0.19	0.36	0.48	0.60
C		0.04	0.08	0.04	0.00	0.00	0.01	0.04	0.28		0.13	0.13	0.15	0.00	0.03	0.11	0.18	0.45
D		0.20	0.30	0.14	0.00	0.04	0.06	0.18	1.00		0.05	0.03	0.04	0.02	0.02	0.04	0.06	0.11
A	*C*	0.87	0.10	0.09	0.66	0.83	0.86	0.93	1.00	*Ec*	0.30	0.19	0.28	0.00	0.16	0.31	0.44	0.64
B		0.43	0.12	0.14	0.23	0.35	0.42	0.50	0.64		0.77	0.11	0.15	0.60	0.69	0.81	0.85	0.92
C		0.22	0.11	0.16	0.07	0.14	0.21	0.29	0.43		0.94	0.07	0.08	0.76	0.91	0.97	0.99	1.00
D		0.06	0.06	0.06	0.00	0.01	0.04	0.08	0.17		0.74	0.25	0.19	0.19	0.72	0.85	0.90	0.95
A	*Ex*	0.85	0.09	0.08	0.65	0.81	0.86	0.90	1.00	*So*	0.94	0.04	0.04	0.86	0.92	0.94	0.96	1.00
B		0.62	0.14	0.14	0.32	0.55	0.60	0.69	0.83		0.67	0.09	0.11	0.52	0.63	0.67	0.73	0.84
C		0.40	0.20	0.28	0.04	0.28	0.45	0.56	0.61		0.36	0.17	0.22	0.08	0.26	0.35	0.48	0.61
D		0.15	0.13	0.20	0.00	0.04	0.11	0.24	0.39		0.17	0.11	0.08	0.00	0.12	0.15	0.20	0.37
A	*SoMean*	0.95	0.03	0.03	0.90	0.93	0.94	0.96	1.00	*SoSecSD*	0.09	0.05	0.05	0.00	0.07	0.10	0.12	0.15
B		0.76	0.10	0.17	0.60	0.69	0.76	0.85	0.89		0.29	0.19	0.26	0.09	0.13	0.21	0.39	0.66
C		0.49	0.21	0.21	0.00	0.41	0.43	0.62	0.77		0.53	0.22	0.35	0.24	0.36	0.50	0.71	1.00
D		0.39	0.17	0.29	0.09	0.25	0.41	0.54	0.59		0.44	0.14	0.21	0.17	0.34	0.44	0.56	0.63
A	*ExMean*	0.86	0.08	0.07	0.68	0.84	0.87	0.91	1.00	*ExSD*	0.24	0.16	0.22	0.00	0.12	0.24	0.35	0.52
B		0.62	0.14	0.13	0.34	0.57	0.61	0.69	0.84		0.46	0.15	0.14	0.20	0.41	0.46	0.55	0.69
C		0.41	0.20	0.26	0.05	0.30	0.46	0.56	0.65		0.77	0.16	0.26	0.49	0.61	0.80	0.87	1.00
D		0.17	0.13	0.19	0.00	0.06	0.15	0.26	0.40		0.58	0.15	0.22	0.36	0.46	0.58	0.68	0.84
A	*FD*	0.01	0.04	0.00	0.00	0.00	0.00	0.00	0.15	*CCseg*	0.01	0.01	0.03	0.00	0.00	0.00	0.03	0.03
B		0.14	0.07	0.13	0.04	0.08	0.13	0.20	0.23		0.20	0.12	0.18	0.03	0.13	0.20	0.31	0.38
C		0.23	0.14	0.20	0.08	0.14	0.19	0.35	0.50		0.55	0.19	0.33	0.30	0.36	0.55	0.69	0.88
D		0.61	0.25	0.40	0.27	0.38	0.63	0.79	1.00		0.78	0.17	0.11	0.35	0.76	0.79	0.87	1.00
A	*CCcut*	0.00	0.01	0.01	0.00	0.00	0.00	0.01	0.01	*Ro*	0.04	0.03	0.03	0.00	0.02	0.04	0.05	0.10
B		0.16	0.11	0.17	0.01	0.07	0.17	0.24	0.32		0.21	0.08	0.09	0.08	0.18	0.21	0.27	0.32
C		0.47	0.22	0.36	0.17	0.26	0.46	0.63	0.83		0.31	0.13	0.13	0.14	0.24	0.27	0.37	0.55
D		0.73	0.21	0.24	0.23	0.60	0.78	0.83	1.00		0.70	0.17	0.20	0.45	0.60	0.74	0.80	1.00
A	*SSD*	0.05	0.05	0.05	0.00	0.01	0.04	0.06	0.17									
B		0.21	0.10	0.14	0.09	0.12	0.21	0.26	0.42									
C		0.39	0.15	0.17	0.19	0.28	0.39	0.45	0.70									
D		0.58	0.22	0.28	0.21	0.43	0.59	0.72	1.00									

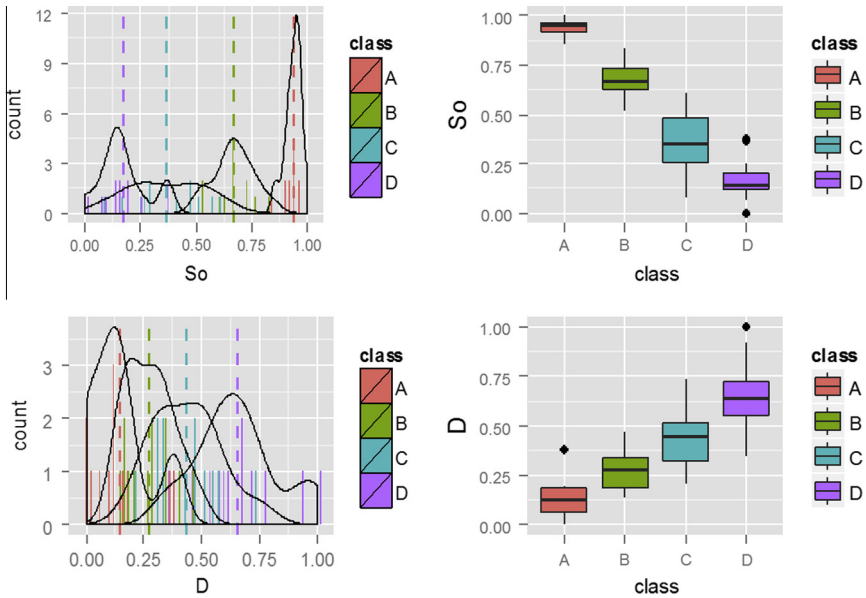

Fig. 6. Exploratory data analysis of homogeneity of variance. With illustrative propose, results are summarize for two variables: *So* does not satisfy the null hypothesis and *D* does and will be rejected. Classes code: A = spherical, B = ellipsoidal, C = linear and D = branched.

of the trees and is modified during training to maximize the model accuracy.

The models are trained using the cross-validation method because its computational cost is not notably high. This method involves dividing the training dataset into 10 subsets. During this process, 9 subsets are selected to train the model, and one resting subset is used to calculate the error of the incomplete sample classification. This procedure is repeated ten times with the testing subset rotated. The average error is calculated as the arithmetic mean of ten errors of the partial samples. The following metrics, which are described in Table 1, are analyzed to optimize the results of the model. Parameter α is also optimized during this training because its value maximizes the model accuracy.

Finally, the best model is tested using a leave-one-out technique, which allows one to test a new data set that was previously split from the original data and not used during the training process. The data in the testing subset are randomly selected. This testing technique provides lower accuracy but more realistic results than the cross-validation method [41].

3. Results and discussion

3.1. Structure data analysis

The basic statistics were calculated and analyzed to obtain a general overview of the variable distributions (Table 2). The differences in performance with regards to the classification capacity are illustrated in Fig. 6. The graphics represent the results of two variables: *So* at the top and *D* at the bottom. The left side shows the non-parametric distributions and the mean values of *So* and *D* grouped in classes. For *So*, the mean values are widely separated, and the variable values of each class are almost split except for class C, whose values are mixed with those that were calculated for aggregates in the B and D classes. The performance for attribute *D* is different: the mean values are closer to each other, and the variable values are mixed for the four classes, so this attribute will most likely not contribute to the classification. On the right side of Fig. 6, the box plot shows the distribution in quartiles of the values, which were calculated for both variables, and, consequently, the ability to distinguish the classes. Evidence of outliers in these two variables is also shown. From the preliminary overview for all data, it was concluded that the use of robust statistics for parametric distributions and statistics for nonparametric distributions would be more adequate in the subsequent multivariate analysis.

3.1.1. Multivariate analysis of the data

3.1.1.1. Analysis of variance. Four statistical tests to analyze the homogeneity of variance were performed for 21 variables in the dataset, and the *p*-values were calculated for all of them. In the used approach, the *p*-value that is calculated for the statistical value is compared to a figure defined by the required level of significance. If the *p*-value is lower than 0.05 (the significance level

Table 3
Results of tests of homogeneity of variance performed for each variable. In bold the variables rejected at this step of the analysis.

ANOVA		Bartlett		Brown–Forsythe		Fligner–Killeen	
Attribute	*p*-value	Attribute	*p*-value	Attribute	*p*-value	Attribute	*p*-value
C	7.81E−23	***D***	**0.2816**	*A*	0.0013	***P***	**0.1367**
Ex	3.44E−14	d_p	5.34E−07	α	0.0015	*Va*	9.56E−07
So	1.85E−20	*SoMean*	6.76E−07			V_p	7.27E−07
ExMean	1.86E−14	***ExSD***	**0.9951**			*n*	0.0034
		CCseg	9.12E−10			*Ec*	0.0352
		CCcut	1.51E−13			*SoSD*	0.0044
		SSD	0.0001			*FD*	2.56E−05
						Ro	0.0082

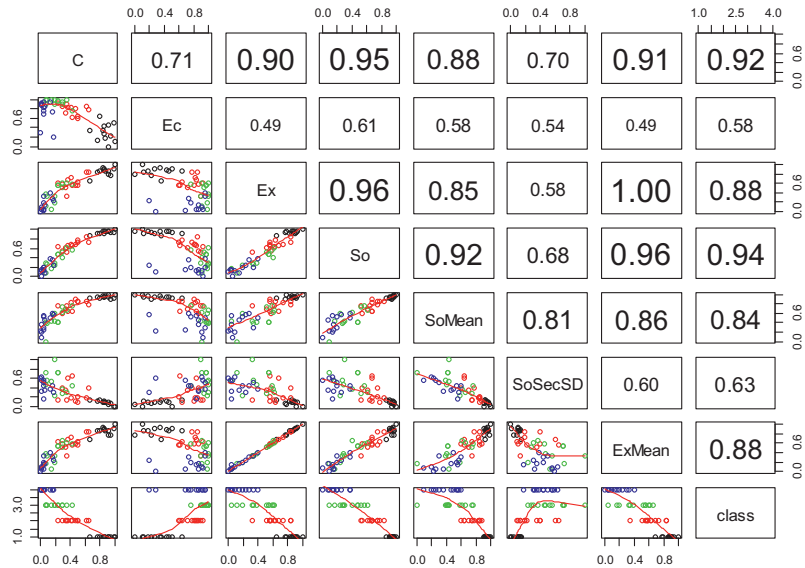

Fig. 7. Scatterplot for variables derived from the aggregate region. The panel on the right side of the diagonal contains the correlation coefficients between variables and the font size is proportional to the level of correlation. The four classes are shown in different colors: spherical = black, ellipsoidal = red, linear = green and branched = blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4
Importance of the first three PC (it is shown only the selected PC, the three first PC).

	PC1	PC2	PC3
Standard deviation	0.97	0.33	0.24
Proportion of variance	0.76	0.09	0.05
Cumulative proportion	0.76	0.85	0.90

of 95% is defined for this work), the variable can distinguish the classes.

The initial analysis of the results showed evidence of contradiction because different tests lead to contradictory conclusions about the null hypothesis assumption. Consequently, it was necessary to assess the individual distribution shapes to select the most appropriate test for each variable. Normal, skewed or extremely heavy-tailed distributions and the existence of outliers were carefully verified in box plots and histograms. Table 3 summarizes the analysis results, showing the chosen test for each variable and the computed *p*-value. When the analysis of variance was complete, the variables *P*, *D*, and *ExSD* were discarded from the dataset for the subsequent analysis.

3.1.1.2. Analysis of correlation. The correlation analysis is used to identify and measure the associations among pairs of variables. The level of collinearity was graphical and numerically studied for all pairs among the remaining eighteen variables. Fig. 7 shows

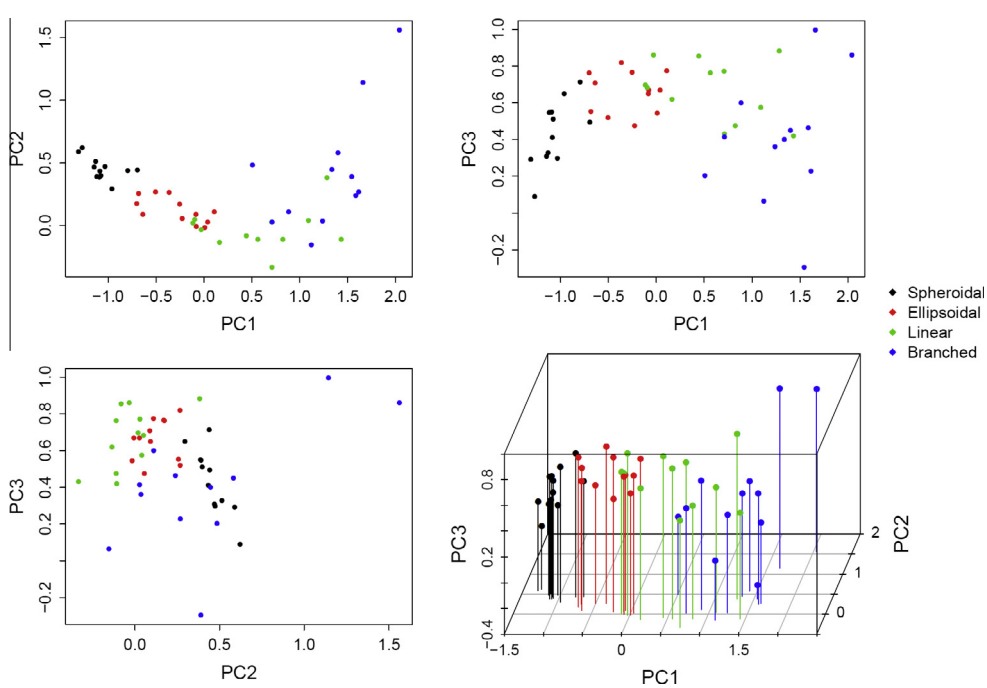

Fig. 8. Exploratory analysis of the selected PC after applying PCA.

the obtained results for the attributes that were extracted from the aggregate region. There is a high level of collinearity for many variable pairs; however, because of the loss of information in the variable discards, only the pair with a correlation factor between 0.90 and 1.00 is further discussed. For all variable pairs with correlation factors in this range, the analysis was individually extended to the four classes. The rejecting criterion was that only if the level of correlation of the four classes was equal or higher than the overall value, the pair of variables was split to decide which variable will be discarded. Because only the pair *ExMean–Ex* satisfied this criterion, one of them should be chosen for rejection. Because in the previous analysis of variance, *ExSD* was discarded and *ExMean* and *ExSD* correspond to the discrete approach of the variable Extent, it was decided to discard the variable *ExSD*. Thus, according to the correlation analysis, the *Ex* discrete approach does not provide additional information to improve the shape classification.

3.1.1.3. Principal component analysis. The main objective of a principal component analysis is to determine the smallest number of required components to reproduce and explain the dependent variable using a set of independent variables. Prior to the analysis, the dataset must be preprocessed (centered and rescaled) to improve the PCA results.

After processing the remaining seventeen variables, the PCA was performed and new principal components (PC) were obtained as lineal combination of the original variables. The analysis results are summarized in Table 4, which shows the part of the total variance contained in the first three PCs. The analyst decides the number of PCs to be considered in a balanced assessment of the explained percentage of variance and the achieved dimensional reduction. In this work, the problem, which was originally defined in twenty-one dimensions and subsequently reduced to seventeen, can be represented by three new dimensions, which are defined by the first three PC, with a 10% loss of information.

The PCA results can be graphically analyzed in two- or three-dimensional plots (Fig. 8). In these new dimensionally reduced spaces, the capacity to split the four morphological classes depends on the percentage of variance that is explained by the PC on the axes. The joint analysis of the graphics show that the PCs allow one to split spheroidal and ellipsoidal aggregates with good results, whereas the linear aggregates are mixed with ellipsoidal and branched ones. There is also evidence of two spurious aggregates that are assigned to the branched class; it has been checked that the spurious aggregates correspond to the largest branched aggregates in Fig. 2, which are agglomerates.

The PCA technique reports additional information of interest, the contribution of each original variable to the new principal components, and the weighting coefficients, which are summarized in Table 5. Note that the two new variables *CCcut* and *CCseg*, which are proposed by the authors, have the highest weights in PC1, which is the first principal component that explains 76% of the overall variance. By integrating the weighting coefficients into the plot for the two principal components (Fig. 9), it is possible to obtain information about the relationships among the original variables and between the variables and the aggregate morphological class. The variables that lie in close directions are highly correlated, as shown in the correlation analysis, and groups of variables that point to a class indicate that these variables more significantly

Table 5
Weighting coefficients of each original variable on the new principal components. Coordinates of the center and rotation matrix to compute PCs.

	Center	PC1	PC2	PC3
A	0.1253	0.1421	0.3871	0.1487
α	0.3369	0.2838	0.0347	0.1500
d_p	0.2131	0.1970	0.3247	0.1631
Va	0.0717	0.1055	0.3894	0.1744
V_p	0.0612	0.0969	0.3791	0.2069
n	0.2796	−0.2317	0.1558	−0.2248
C	0.3931	−0.3139	0.1859	−0.0702
Ec	0.6911	0.1840	−0.3554	0.7156
Ex	0.5040	−0.2827	0.0707	0.2583
So	0.5362	−0.3134	0.1227	0.1017
SoMean	0.6463	−0.2512	0.1261	0.0909
SoSecSD	0.3368	0.1650	−0.2811	0.0092
FD	0.2475	0.2420	0.3087	−0.0818
CCseg	0.3848	0.3284	−0.0980	−0.2116
CCcut	0.3411	0.3157	−0.0592	−0.2566
Ro	0.3167	0.2576	0.1068	−0.2692
SSD	0.3054	0.2294	0.1671	0.1121

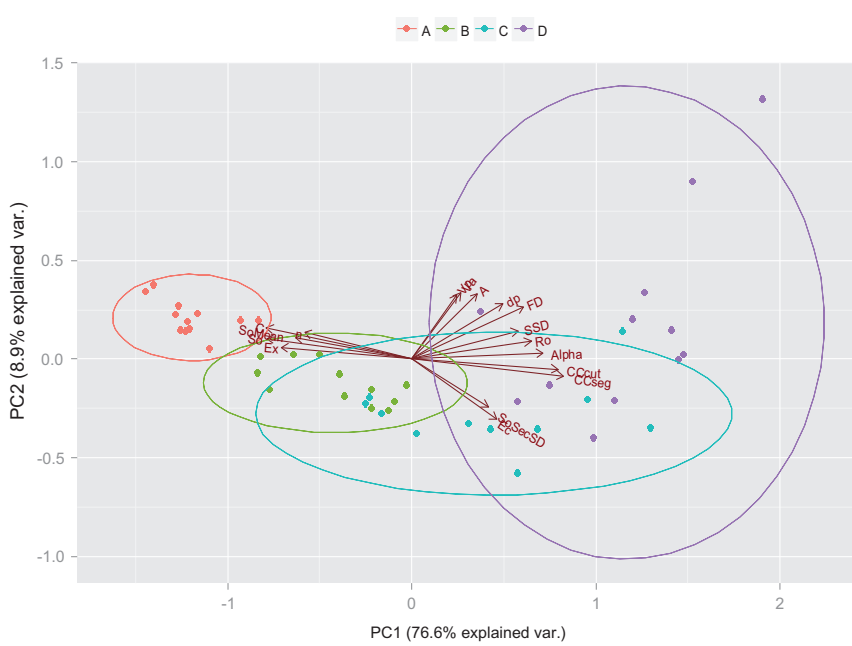

Fig. 9. PC1 and PC2 biplot. Where class is A = spheroidal, B = ellipsoidal, C = linear and D = branched.

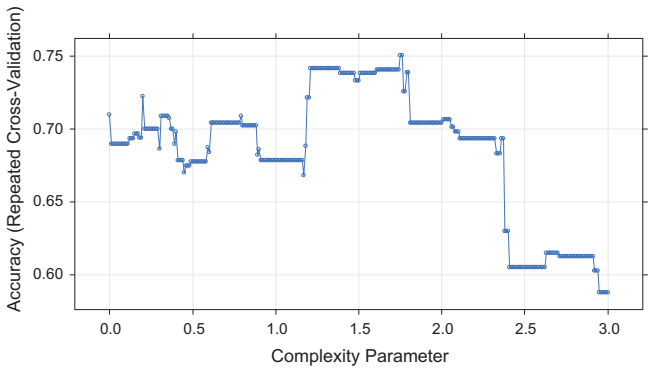

Fig. 10. Variation of the accuracy on the selected classification model related with the complexity of the tree (α) for fixed values of all other parameters in the algorithm (minbucket = 1, minsplit = 2, maxdepth = 9, niterations = 10,000 and ntrees = 100 and probability of 33% for crossover operator, 33% for minor/major split rule mutation (50% per each) and 33% for split/prune operator (50% for each)).

Table 6
Evtree algorithm parameters used to build the most accurate tree.

Population parameters	Value	Evolutionary learning parameters	Value
Minbucket	1	Pcrossover	0.2
Minsplit	2	Psplit	0.2
Maxdepth	9	Pprune	0.2
Niterations	10,000		
Ntrees	100		
α	1.75		

affect this class. The weighting coefficients allow one to compute the PC values for a new aggregate, which is preceded by the transformation of the coordinates to the PC-based system; by drawing the new point into the plots, the aggregate can be assigned to a class. The method does not report any information on the success/failure probability. The colored ellipses in the graphic represent the 95% confidence level in finding the point that represents a new aggregate in the class.

To summarize the results of the multivariate analysis of the data, it can be concluded that the original twenty-one attributes of the aggregates can be reduced to seventeen with acceptable loss of information. The problem dimension can be dramatically reduced to only three principal components if only the 90% variance is explained. Then, the aggregate morphology is assigned to one of the four proposed classes by applying the PC computations to the geometrical and morphological characteristics, which are obtained from its TEM image.

3.1.2. Classification tree

In the last step of the method, a simple aggregate categorization model was constructed based on the previous statistical analysis. Thus, a classification model based on decision trees was implemented with the three principal components as variables.

The method in this part of the work requires the splitting of 70% of the data to train and validate the models. The remaining 30% is used for the final testing of the classification models, in a leave-one-out technique. During the training process, several parameters of the algorithm must be varied to maximize the model accuracy because the optimal choice of these parameters depends on the particular dataset. The models were validated using the 10-fold cross-validation technique, and the complete process was repeated ten times for each model.

Two types of operators are varied until the accuracy is maximized:

- The first type of operator is related to the population of the dataset, which considers parameter such as the minimum sum of weights at a terminal node (minbucket), the minimum sum of weights at a node to be considered for splitting (minsplit), the maximum depth of the tree (maxdepth), the number of iteration in case the process does not converge (niterations), the number of trees (ntrees) and the complexity of the tree (α).
- The second type of operator is related to the evolutionary learning process, which considers the probability of using one type of mutation operators or the crossover operator.

Fig. 10 illustrates the accuracy variation of the model with α for fixed values of all other parameters. The best tree that was selected was built with the parameters in Table 6. The maximum achieved accuracy in cross-validation was 75.1% (Fig. 10).

The evtree can identify the most important attributes to select the output variable and discern those that do not affect the process. In the most accurate tree, the selected variables were PC1 and PC2; PC3 did not appear in the tree. Another interesting characteristic of the model is that only four leaves are required to build the tree, which produces a structure that is simple to explain using only

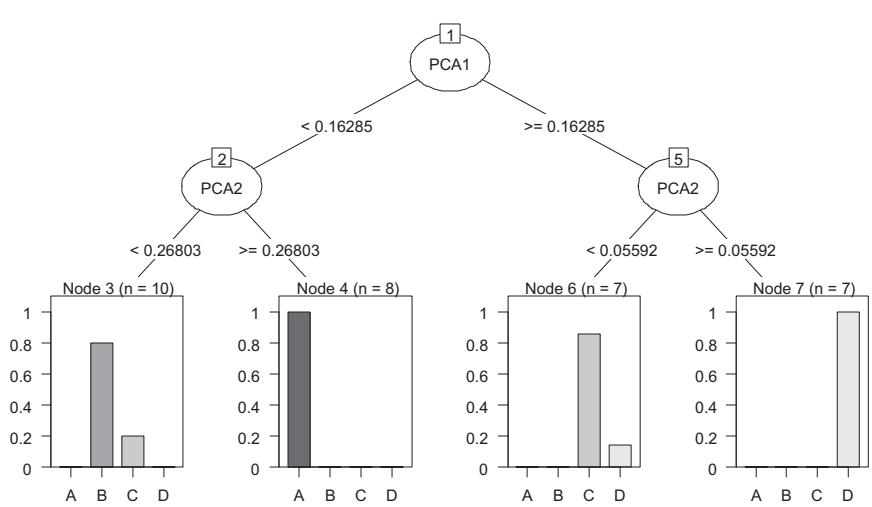

Fig. 11. Classification tree that assigns carbon black aggregates to the four studied classes. A = spherical, B = ellipsoidal, C = linear and D = branched.

Table 7
Testing confusion matrix and classifier performance.

		Spherical	Ellipsoidal	Linear		Branched
Confusion matrix	Spherical	4	0	0		0
	Ellipsoidal	0	4	0		0
	Linear	0	1	2		1
	Branched	0	0	2		2
		TP Rate	FP Rate	Precision	Recall	FMeasure
Classifier performance	Spherical	1.0	0.000	1.000	1.0	1.000
	Ellipsoidal	1.0	0.083	0.800	1.0	0.889
	Linear	0.5	0.167	0.500	0.5	0.500
	Branched	0.5	0.083	0.667	0.5	0.572

two principal components (Fig. 11). A new aggregate with defined values of PC1 and PC2 can be classified according to the threshold values of these two variables.

Finally, the best model obtained during the training step was tested with the data subset that was split at the beginning and not used in the training process in the leave-one-out technique. The aggregate classification results are shown in Table 7, where the confusion matrix and the classifier performance are summarized. The sixteen aggregates (four per class) that were used to test the model were classified with a global accuracy of 75%, improving slightly results shown in [13]. Note that this 75% accuracy is obtained using only the first two PC that contain 85% of the explanation of the total variance. Both spheroidal and ellipsoidal aggregates were 100% correctly classified. The linear aggregates were 50% correctly assigned to their class, 25% were assigned to the ellipsoidal class, and 25% were assigned to the branched classes. The branched aggregates were 50% correctly classified, and the other 50% was assigned to linear class. Thus, within an acceptable global accuracy of the model, the linear and branched aggregates are the most difficult to correctly assign to their class since their shapes are quite similar.

Because the final classification tree uses only two variables (PC), it can be plot in a bi-dimensional graph (Fig. 12). The tree splits the dataset space into four different regions according to the classes. The locations of the data during the training and testing steps show the successes and failures in classification that were achieved by the model. Besides, Fig. 12 shows that misclassification made with linear and branched classes is for a very small margin, confirming the usefulness of this methodology.

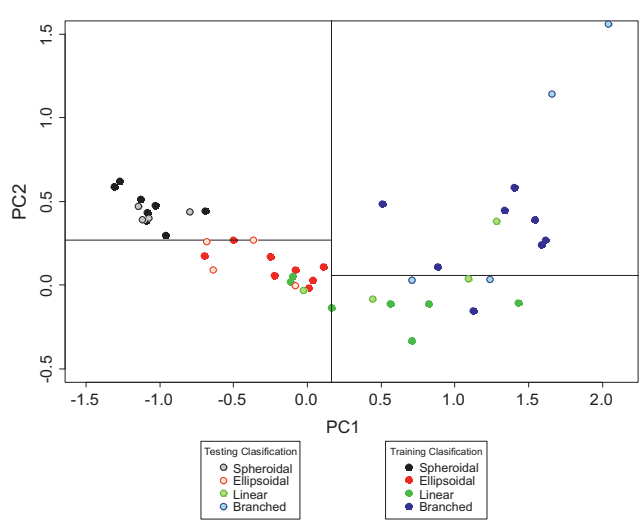

Fig. 12. Results obtained by the classification tree. The tree splits the space according to the four classes and the training and testing data are classified by the model.

4. Conclusions

The morphology of the reinforcing elements in composite materials has a significant effect on their final properties, and yet the efficient quantification of the occurrence of different morphologies and the ratio among them remain unsolved. Such is the case of carbon black – which has been used here as study case – when is used as reinforcing agent in polymer systems.

In this work, a new three-step method is proposed and successfully tested to assess the characteristics of the aggregates and classify them into the four mostly accepted shapes: spheroidal, ellipsoidal, linear and branched.

First, a multivariate analysis is performed to achieve the most appropriate dataset to solve the classification problem. After measuring morphological and geometrical values (obtaining 21 features) for 48 elements using TEM images, the significance of each feature to predict the class is successfully identified and used to reduce the number of features by discarding the non-significant ones (keeping 17 features).

Then, the problem of high dimensionality is solved using principal component analysis, obtaining a new set of three uncorrelated principal components that retain the 90% overall-variance of the data (generating and keeping just 3 new features).

Finally, the classification is performed using decision trees based on evolutionary algorithms, technique that clearly explains the data structure and provides a highly intuitive insight into the interactions among the variables, obtaining an overall accuracy of 75%, and a notably easy to use and interpret tree formed only by four leaves and two threshold values (one for PC1 and one for PC2).

To sum up, the main contribution of this paper is to propose a successful method for characterizing the carbon black filler and classifying the aggregates into specific morphological categories. Thus, it is possible to determinate the shape of these particles, which could be consequently used to obtain a better understanding of how the filler affects the final properties of the composite material.

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